BIOT6900 – Module 2 Assignment
Multi-Omics Target Discovery in Pancreatic Ductal Adenocarcinoma (PDAC)

1. Disease and Data Choice

I chose PDAC for this assignment because it has one of the poorest five-year survival rates of any solid tumor, and while its core driver mutations (KRAS, TP53, SMAD4, CDKN2A) are well characterized, the disease remains substantially under-drugged relative to how well its genetics are understood. The gap between known biology and available therapies makes it a good candidate for a target discovery experiment. The literature provides a clear, evidence-based set of expected hits to validate the pipeline's results.

All three data layers came from the CPTAC-PDAC cohort, downloaded from LinkedOmicsKB, part of the CPTAC pan-cancer Data Freeze v1.2. Every file used was open access, no data-use agreement was required. The five files retrieved were the **RNA-seq gene-level matrices for tumor and normal samples, the proteomics gene-level abundance matrices for tumor and normal samples, and the gene-level binary somatic mutation matrix**.

CPTAC is useful because the tumor samples are matched across RNA, protein, and mutation data, meaning that the same group of around 140 patients was analyzed at all three levels. This makes it possible to directly compare RNA and protein measurements from the same patients, like we did in the Part 1 breast cancer example.

For this assignment, I took a slightly different approach. Instead of comparing RNA and protein levels within the same tumor patients, I compared the tumor samples with the available normal samples. There were 21 normal samples for RNA and 75 for protein, so these groups were not matched to the tumor patients or to each other. Because of this, I integrated the data at the gene level rather than the sample level. I matched the datasets using common gene identifiers and then compared the results for each gene across the RNA and protein layers.

I chose this method because the tumor-versus-normal comparison gives a more meaningful differential signal for each gene. For every gene, I could calculate a log2 fold change and a p-value, instead of only ranking genes based on their raw abundance. For the genomic layer, instead of using a GWAS association statistic, I calculated the percentage of tumor samples that had a somatic mutation in each gene using the binary mutation matrix. This is different from a GWAS p-value because it represents how frequently a gene is mutated within the tumors, rather than its association with inherited disease risk. So, I used mutation frequency as an alternative source of genomic evidence and did not treat it as actual GWAS data.

2. Top Targets

After matching all three datasets using the same version-stripped Ensembl gene IDs, **1,880 genes** were shared across all three layers. The RNA dataset originally contained 60,579 genes and the protein dataset contained 11,057 genes. The mutation dataset was much smaller, with only 3,362 genes because it only included genes that had at least one observed mutation. Because of this, the mutation data was the main factor limiting the number of genes included in the final analysis.

To score the genes, I gave equal weight to all three layers, following the approach used in the course. The score included the absolute RNA log2 fold-change, absolute protein log2 fold-change, and mutation frequency. Since these measurements are on different scales, I first converted each one into a 0–1 rank percentile and then added them together to get the final score.

RNA and protein both provide differential-expression evidence, while mutation frequency can provide additional information about genes that are commonly altered in tumors. However, the mutation data also has more limited gene coverage. Giving it a higher weight could therefore favor genes simply because mutation information was available for them, so equal weighting was the more reasonable approach for this analysis.

Top 10 Candidate Genes from Multi-Omics Integration

The top 10 genes based on the final score were **MUC16, CFTR, COL11A1, CEL, LRP1B, BRSK2, ACADL, NOS1, XDH, and KRT19**. Although these are not the typical driver genes commonly associated with PDAC, some of them still make biological sense. For example, MUC16 (CA-125) is a known tumor marker, while KRT19 is associated with pancreatic ductal cells and is commonly expressed in PDAC. COL11A1 has also been linked to the stromal and fibrotic environment seen in pancreatic cancer.

One interesting result was CFTR, which showed strong downregulation. CFTR is related to normal pancreatic duct cell function, and changes in its expression have been reported in pancreatic cancer. However, it is not usually considered a major PDAC target. Because of this, I would treat CFTR as an interesting gene for further investigation rather than immediately considering it a strong therapeutic target. Its relevance could be checked using a curated database such as Open Targets.

Gene	RNA log2FC	Protein log2FC	Mut. freq.	Concordant	Score
MUC16	3.696	1.117	0.057	True	0.979
CFTR	-2.438	-1.819	0.029	True	0.978
COL11A1	2.417	1.402	0.029	True	0.973
CEL	-6.348	-2.550	0.019	True	0.962
LRP1B	-1.561	-1.060	0.086	True	0.957
BRSK2	-3.083	-1.955	0.019	True	0.956
ACADL	-3.025	-1.983	0.019	True	0.956
NOS1	-2.205	-0.978	0.029	True	0.955
XDH	2.788	0.836	0.029	True	0.952
KRT19	2.688	1.433	0.019	True	0.950

Recovery of known PDAC drivers

None of the four major PDAC driver genes (**KRAS, TP53, SMAD4, and CDKN2A**) appeared in the top 15 genes from the overall score. Instead of assuming they were completely missed, I checked where these known drivers ranked in the full dataset.

Rank	Gene	RNA log2FC	Protein log2FC	Mut. freq.	Note
32	CDKN2A	1.206	0.601	0.229	Concordant; recovered near the top of the full ranking
69	KRAS	0.685	0.414	0.962	Near-universal mutation frequency; modest RNA/protein shift
381	TP53	-0.046	0.803	0.829	Discordant — see Section 3
398	ARID1A	-0.370	-0.193	0.057	Weak signal across all three layers
530	SMAD4	-0.727	-0.017	0.190	Weak protein signal despite RNA/mutation evidence
—	RNF43	present	MISSING	present	Dropped by the inner join — absent from the proteomics file

KRAS is especially interesting in this comparison, its mutation frequency was 0.962, meaning that about 96% of the tumors had a KRAS mutation. However, its RNA and protein fold-changes were much smaller, at 0.685 and 0.414. This makes sense because KRAS-driven PDAC is mainly associated with activating mutations rather than simply producing much more KRAS RNA or protein. Therefore, looking only at RNA or protein expression would not fully capture the importance of KRAS.

This result also shows why combining different omics layers can be useful. The mutation data clearly captures the importance of KRAS, even though its expression changes are relatively modest. **CDKN2A**, on the other hand, shows evidence across all three layers and therefore ranks much

higher overall at rank 32. Overall, these results show that known cancer drivers can have very different patterns across genomic, transcriptomic, and proteomic data, and may not necessarily appear among the highest-ranked genes when all three layers are weighted equally.

3. Understanding a Discordant Gene

None of the top 15 genes showed clear discordance, where RNA and protein changed in opposite directions or showed very different patterns. So, I looked further down the ranking for a gene with a clear biological explanation. **TP53**, ranked 381, was a good example. Its RNA log2 fold-change was almost unchanged at -0.046, while its protein log2 fold-change was much higher at 0.803. it also had a high mutation frequency of 0.829.

There is a biological reason why TP53 can show this pattern. Many TP53 mutations in cancer are missense mutations that produce a stable mutant p53 protein. Normally, wild-type p53 is kept at low levels because it is rapidly degraded through pathways involving MDM2. However, mutant p53 proteins can become stable and accumulate inside tumor cells. Because this happens mainly at the protein level, TP53 protein can increase without a similar increase in its mRNA.

This helps explain why TP53 shows almost no RNA change but a noticeable increase in protein abundance in this dataset. Instead of simply treating the difference between RNA and protein as noise, it may reflect the effect of TP53 mutations on protein stability. I chose TP53 for this example because it clearly shows how differences between omics layers can provide useful biological information that may not be visible when looking at only one layer.

4. Limitations and Interpretations of the Analysis

- **Unmatched samples:** The tumor and normal samples came from **different individuals**, and the RNA and protein datasets also had different numbers of normal samples. Therefore, the analysis compares **average tumor-vs-normal differences at the gene level** rather than tracking RNA and protein changes within the same patients. This means I can identify general patterns across groups, but I cannot claim that an RNA change and a protein change occurred together in the same individual.

- **Different gene coverage:** Only **1,880 genes** were shared across all three datasets. Important genes such as **RNF43** were lost because they were missing from one of the layers.
- **Mutation frequency is not GWAS data:** It shows how often a gene is mutated in the tumor samples, not whether the gene is associated with inherited disease risk.

- **PDAC has high stromal content:** Bulk RNA and protein data contain signals from both tumor and surrounding stromal cells, so the measurements do not represent pure tumor-cell biology.
- **Overall:** This analysis is useful for identifying candidate genes and comparing evidence across multiple omics layers, but matched samples or single-cell data would provide stronger and more specific conclusions.

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